

Lucas Doyle

Senior Machine Learning Engineer | San Francisco, CA



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Sounds cliché, but making software do something in the real world is what excites me. I specialize in zero-to-one problem solving, building prototypes that evolve into scalable systems, and shipping robotics and ML products that operate reliably in the field. Experience spans robotic arms, drones, autonomous systems, and embedded ML at scale.

Experience

Medra - Team Lead, AI Scientist

San Francisco, CA | 2026 - Present

Python, TypeScript (React), Go, C# (.NET), Claude agents

Autonomous Robotic Laboratory & AI Agents

- Lead the AI scientist pod at an AI-run robotic lab startup; DRI for the AI experimentalist on a major pharma customer engagement; co-designed the lab's three-tier software architecture.
- Built Medra OS: fully unattended bare-metal provisioning that scaled the lab from 10 to 100+ robotic workstations; absorbed an overnight pivot to consumer mini PCs, standing up 60 decks in 4 days.
- Integrated lab instruments (thermocycler, qPCR, plate-handoff rail electronics and firmware) in days each, then packaged the process as an agent skill teammates used to bring up more instruments unaided.
- Created a company-wide autonomous SWE agent used by 50+ engineers; most of its work is self-initiated (PR triage, fleet-alarm first response, bug-report dedup) and most automations on it were built by others.
- Own the monorepo developer environment (4 languages) and an internal apps hub hosting 11 apps in regular use across the company.

Samsara - Senior Machine Learning Engineer II

San Francisco, CA | 2018 - 2026

Python, Golang, Typescript (React, Redux, React Native), Android (Java, Kotlin)

Edge ML Infrastructure

- Led end-to-end design and deployment of embedded video ML pipelines powering Samsara's AI dashcams (2M+ devices in the field).
- Shipped safety-critical features (e.g. tailgating, rolling stop, lane departure) running on resource-constrained hardware.

- Built QA device farms, shadow testing systems, and firmware infrastructure for rapid iteration of on-device ML.
- Spearheaded offline evaluation pipelines with multimodal inputs (video, IMU, GPS, hardware-in-the-loop simulation / emulation).
- Designed debugging and telemetry tools to improve observability across firmware and cloud systems.

Airware - Senior Software Engineer

San Francisco, CA | 2014 - 2018

JavaScript (React, Redux, Leaflet, React Native), Python, Swift, PHP, Golang

Drone Platforms & Aerial Robotics

- Built core technologies for autonomous drone systems used in insurance and mining industries.
- Developed applications for configuring and flashing avionics on fixed-wing and multirotor drones.
- Built geospatial data processing pipelines and survey-grade mapping workflows for large-scale aerial datasets.
- Contributed to system-level integrations during M&A, driving engineering alignment across platforms.

MUJIN Inc. (株式会社MUJIN) - Software Engineer

Tokyo, Japan | 2012 - 2014

Python (django, OpenRAVE), JavaScript (Backbone/Marionette, WebGL)

Industrial Robotics

- Early engineer at a Japanese robotics startup, developing the MUJIN Controller for robotic arm workcell planning and optimization.
- Built real-time customer UIs with WebGL and WebSockets for bin-picking and industrial automation systems.
- Collaborated with multinational robotics engineers and deployed solutions for clients including Canon and Honda.

Education

Rensselaer Polytechnic Institute

B.S. Computer and Systems Engineering | 2008 - 2012

Rensselaer Center for Open Source, Red & White Alumni Organization

Boeing Robotic Wingbox

Capstone - simulated intra-wing robot for aircraft inspection. ([project link](#))

Patents

Lane departure monitoring ([US12165393B1](#))

Context based action menu ([US12150007B1](#))

Unmanned aerial vehicle privacy controls ([US20220392353A1](#))

Miscellaneous

- Mechanical design / CAD / 3d printing
- Electronics / microcontrollers / remote control / robotics
- Homelab / VPN / LocalLlama / extensive self-hosting experience
- Occasional paid and pro-bono consulting
- Cat lover, marathon runner (BQ 2023), mixologist, climbing